

Residual $SO(4, 2)$ Cohomology of Free Weyl Gravity on the Conformal Cylinder: Cartan Homotopy, Weyl-Square Classes, and Their Residual Pairing

Research-programme draft
Authorship and attribution to be finalized after external review

Draft for external review, July 2026

Abstract

Treating all fifteen cylinder conformal transformations, including compact time translation D , as residual gauge transformations, we compute a minimal absolute $SO(4, 2)$ Chevalley–Eilenberg complex for free four-dimensional Weyl gravity on $\mathbb{R} \times S^3$. Its coefficient module is the parity-complete on-shell Weyl-curvature Fock module. The standard invariant one-particle form is indefinite, with cylinder signs $+I_E \oplus (-I_A) \oplus (-I_L)$.

The key mechanism is the Cartan homotopy formula. The residual differential obeys $d\iota_D + \iota_D d = \mathcal{L}_D$, so every subcomplex of nonzero total compact degree is contractible. In the Hamada-centered four-ghost polarization, the selected absolute degree has ghost compact degree bounded below by -4 , and only matter weights at most four can contribute. The vacuum and one-particle sectors are acyclic, while a cutoff-complete absolute calculation in the lowest two-particle sector gives

$$H_{\text{res}, \min}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.$$

Here $G_{\text{res}} = I_2$ is the induced Hermitian complementary-degree CE pairing in the displayed residual basis. The survivors are ghost-dressed vertex or deformation classes, not one-particle graviton states. Descent identifies their parity-even and parity-odd combinations with $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ and $C_{\mu\nu\rho\sigma}\tilde{C}^{\mu\nu\rho\sigma}$. The first varies to the Bach equation; the second is the Pontryagin boundary direction. After quotienting variationally trivial bulk functionals, the local dynamical deformation space is one-dimensional and has residual Gram matrix I_1 .

This algebraic residual theorem is exact once the on-shell Weyl module, the Hamada ghost polarization, and full residual conformal gauging are assumed. The last assumption is a physical choice: if D is retained as a Hamiltonian or boundary charge, the contraction is not a physical quotient. The flat adjoint BGG resolution and the topology of $\mathbb{R} \times S^3$ additionally give an all-level smooth isomorphism between the Bach quotient and a geometrically defined on-shell curvature system. An exact symbolic cylinder construction supplies finite-mode metric preimages for every $E/A/L$ block and identifies that geometric system with the algebraic positive-energy module. An end-to-end polynomial calculation additionally constructs the metric, ghost, equation-antifield and identity-antifield rows, measures the noncompact homotopy defects, transfers the strict residual action, and reproduces the centered cohomology without importing the hand-specified $E/A/L$ matrices. Two identifications remain conditional: the cross-energy cyclic local-BV pairing and its normalization, and derivation of the centered BFV polarization from the chosen closed-universe reduction. Any analytic completion requires additional control. We claim neither a positive graviton Fock space nor quantum anomaly freedom.

Contents

1	Introduction	3
2	The free local pure-Weyl complex	5
2.1	Gauge, curvature, and equation maps	5
2.2	The global four-dimensional deformation complex	6
2.3	The Bach quotient as an on-shell curvature space	7
2.4	The finite-jet calculation as an independent convention audit	9
2.5	The finite global zero-mode pair	10
2.6	Domain boundary of the theorem	10
3	The on-shell Weyl module	11
3.1	Character and cylinder towers	11
3.2	Invariant form and exact generator action	11
4	Hamiltonian moment-map normalization	12
4.1	The conformal Taub constraint	12
5	Residual BRST and Cartan homotopy reduction	14
5.1	The absolute residual complex	14
5.2	Full residual gauging is a physical assumption	14
5.3	Cartan homotopy	15
5.4	Finite centered inventory	15
6	Complete centered cohomology	16
6.1	The relative weight-four kernel	16
6.2	Absolute kernel and image	17
6.3	The canonical residual CE pairing	17
7	Descent and the dynamical/topological split	19
7.1	Why weight four is selected	19
7.2	Parity basis	19
7.3	Euler–Lagrange quotient	20
8	The local-to-residual transfer	20
8.1	The free Fock lift is formal	21
8.2	Equivariant Cartan transfer	21
8.3	Cyclic transfer and the pairing	21
8.4	Raw polynomial instantiation and measured noncompact defects	22
8.5	The filtered local-to-residual complex	23
8.6	End-to-end centered integration in the polynomial category	24
8.7	Two remaining field-theoretic hypotheses	25
9	Scope, interpretation, and open questions	26
10	Conclusion	27
A	Conventions	27

1 Introduction

Pure conformal gravity in four dimensions is defined, up to an overall coupling and the Euler density, by the Weyl-square action

$$S_W[g] = -\alpha_g \int_M d^4x \sqrt{-g} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}. \quad (1)$$

Its fourth-order linearized equation has more solutions than the Einstein equation, and a conventional local reduction produces an indefinite oscillator form. This familiar observation does not by itself determine the state space after all gauge constraints have been imposed. On a compact spatial slice, background conformal symmetries are reducibilities of the local $\text{Diff} \times \text{Weyl}$ gauge system. Treating those residual transformations as gauge introduces a second, finite-dimensional reduction which is absent from a naive oscillator count. It is also a substantive physical choice. In a Dirac or background-independence interpretation of a closed universe, all residual conformal transformations can be quotiented. In a fixed cylinder conformal field theory, by contrast, D is ordinarily the Hamiltonian and the conformal group acts globally. This paper computes the first choice; it does not assert that strict pure-Weyl gravity uniquely selects it.

The organizing question is therefore more basic than a sign assignment: before asking whether a gravitational mode has positive or negative norm, one must first determine whether it survives the full gauge reduction being imposed. The exact answer below concerns the residual reduction. The smooth metric-to-curvature bridge is proved globally, and an exact all-energy cylinder construction identifies its algebraic positive-energy part with the $E/A/L$ module used below. The passage to the complete local-plus-residual BV/BFV theory remains conditional.

The present paper isolates that free classical problem on

$$M = \mathbb{R} \times S^3, \quad \bar{g} = -dt^2 + d\Omega_3^2, \quad (2)$$

with unit cylinder radius. Three spaces must be distinguished:

$$\begin{aligned} &\text{local oscillator cohomology} \supseteq \text{classically integrable perturbations,} \\ &\text{classically integrable perturbations} \longrightarrow \text{residual-BRST classes.} \end{aligned} \quad (3)$$

The first contains the familiar Einstein, vector-descendant, and logarithmic or higher-derivative towers. The last is obtained only after the residual $SO(4,2)$ constraints and their ghosts are included. A linearized oscillator need not survive this final reduction.

Our result is most naturally stated in the language of residual vertex cohomology. The Hamada ghost polarization supplies a top-degree four-ghost factor of compact degree -4 . It pairs with scalar matter primaries of weight four. The complete centered residual cohomology consists of two chiral Weyl-square classes, and their normalized residual CE Gram matrix is positive. Their parity combinations are the dynamical Weyl-square density and the topological Pontryagin density. Thus the positive matrix is attached to classical vertex or theory-space directions. It is not a positive norm on a propagating graviton Fock space; indeed, the one-particle residual cohomology vanishes.

Relation to previous work. The conformal deformation detour complex on obstruction-flat backgrounds was constructed by Branson and Gover [1]; its formal self-adjointness and related Yang–Mills detour constructions were developed in [3]. Gover and Peterson give the oriented four-dimensional deformation sequence, while Čap proves that the flat BGG sequence is a fine resolution and computes the same cohomology as the flat twisted de Rham complex [2, 5]. Cylinder harmonics, oscillator charges, and primary building blocks were developed by Hamada and Horata and by Hamada [8, 9]. In the broader conformal-mode/Riegert programme, Hamada identified the residual diffeomorphisms with cylinder conformal transformations [10], constructed the fifteen-generator residual BRST charge, used the product of four proper-conformal ghosts as the centered ghost vacuum, derived the scalar weight-four condition, and displayed a Weyl-square BRST state [11]. None of those ingredients is claimed as new here. We use the harmonic and residual-algebra conventions, but we do not import the Riegert field or the quantum claims of that model. The Weyl-curvature module and its character are consistent with the general conformal higher-spin description and partition-function literature [12, 13]. Finally, the identification of C^2 as the unique local four-derivative self-interaction of one linear conformal graviton, under the assumptions stated there, is due to Boulanger and Henneaux [14].

The defensible new result is narrower and more concrete. Starting with the parity-complete pure-Weyl coefficient module, we compute the complete minimal *absolute* residual complex in the centered degree. We prove that the vacuum and one-particle sectors vanish, that precisely two two-particle classes survive after the incoming-image test, and that their normalized Hermitian residual CE Gram matrix is I_2 . The standard Cartan homotopy makes the particle-number inventory exhaustive, and the Euler–Lagrange map splits the two positive directions into one dynamical and one topological class.

Claim boundary. Four statements must remain separate.

1. The *algebraic residual theorem* is exact for the minimal absolute $SO(4, 2)$ Chevalley–Eilenberg complex built on the on-shell Weyl module, the stated centered ghost polarization, and full residual conformal gauging.
2. The smooth complexified metric-to-geometric-curvature bridge is an exact global theorem. Its restriction to the D -finite cylinder category is also exact: the explicit $E/A/L$ harmonics give same-energy metric preimages, nonzero curvature intertwiners, and character exhaustion. Neither theorem is a statement about a completed Hilbert/Krein domain.
3. In the finite polynomial conformal category, an exact metric-BV calculation now includes the ghost and antifield detour rows, constructs a homotopy-equivariant retract, transfers the strict residual action, and reproduces the centered kernel and image. Identifying its representative form with the complete cross-energy local-BV pairing remains conditional, as does the residual BFV boundary choice which induces the centered ghost insertion and its normalization. These remaining hypotheses are displayed rather than hidden.
4. Calling the resulting classes physical states additionally requires a boundary and gauge principle which makes all fifteen conformal generators, including D , gauge rather than global charges. We call them *candidate physical classes under full residual conformal gauging*.

No statement in this paper establishes quantum BRST nilpotency, anomaly cancellation, interacting pseudo-unitarity, or a positive asymptotic particle space.

The local-to-residual bridge belongs to the general subject of local BRST cohomology with antifields, gauge fixing, and descent [15, 17]. Metsaev's ordinary-derivative BRST–BV construction gives a concrete realization of conformal fields and their conformal algebra on fields and antifields [18]. Dneprov and Grigoriev construct a minimal nonlinear Weyl-jet model and compatible presymplectic structure [19]. Their curvature-dependent ghost transformations make that model a valuable comparison, but not an immediate proof that the *free* transferred residual differential is strict. Neither work supplies the global cylinder zero-mode normalization used here.

2 The free local pure-Weyl complex

2.1 Gauge, curvature, and equation maps

Let $h_{\mu\nu}$ be a metric perturbation and let (ξ^μ, σ) be an infinitesimal Diff $\times$ Weyl parameter. The full linear gauge map is

$$K(\xi, \sigma)_{\mu\nu} = \mathcal{L}_\xi \bar{g}_{\mu\nu} + 2\sigma \bar{g}_{\mu\nu} = 2\bar{\nabla}_{(\mu} \xi_{\nu)} + 2\sigma \bar{g}_{\mu\nu}. \quad (4)$$

After quotienting the trace, it becomes the conformal Killing operator

$$(K_0 \xi)_{\mu\nu} = 2\bar{\nabla}_{(\mu} \xi_{\nu)} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{\nabla} \cdot \xi. \quad (5)$$

Let

$$C_1 h = \left. \frac{d}{d\varepsilon} C[\bar{g} + \varepsilon h] \right|_{\varepsilon=0} \quad (6)$$

be the linearized Weyl-curvature map, and let B_{lin} denote the action-normalized linearized Bach operator. Formal adjoints use the spacetime action pairing on a domain for which temporal and spatial boundary terms vanish.

The conformal cylinder is conformally flat. Naturality and conformal covariance therefore give the exact local identity

$$\boxed{C_1 K = 0} \quad (7)$$

for arbitrary local gauge parameters, not only for the fifteen residual conformal Killing parameters.

Proposition 2.1 (Action-normalized factorization). *Use the repository normalization*

$$S_{\text{red}}[g] = \int \sqrt{-g} \left(R_{\mu\nu} R^{\mu\nu} - \frac{1}{3} R^2 \right) = \frac{1}{2} \int \sqrt{-g} (C^2 - E_4). \quad (8)$$

On a conformally flat background, modulo the Euler boundary term,

$$\boxed{B_{\text{lin}} = C_1^\# C_1}. \quad (9)$$

Consequently,

$$B_{\text{lin}} K = 0, \quad K^\# B_{\text{lin}} = 0, \quad B_{\text{lin}}^\# = B_{\text{lin}}. \quad (10)$$

Proof. Since $C[\bar{g}] = 0$, the mixed second variation of (8) is

$$\delta_h \delta_k S_{\text{red}} = \langle C_1 h, C_1 k \rangle_{\mathcal{W}}. \quad (11)$$

The definition $\delta S_{\text{red}} = \langle h, \mathcal{E}(g) \rangle$ and $B_{\text{lin}} = D\mathcal{E}|_{\bar{g}}$ gives $\langle h, B_{\text{lin}} k \rangle = \langle C_1 h, C_1 k \rangle_{\mathcal{W}}$, proving (9). Equation (7) then gives the first detour identity, formal self-adjointness gives the second, and Hessian symmetry gives the third. $\square$

The Weyl-fibre pairing in Lorentz signature is indefinite. Thus $C_1^\sharp C_1$ is a formal-adjoint factorization, not a positivity factorization. In trace-free conformal geometry the corresponding local sequence is

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2 T^*) \xrightarrow{B_{\text{lin}}} \Gamma(S_0^2 T^*) \xrightarrow{K_0^\sharp} \Gamma(T^*). \quad (12)$$

On Bach-flat backgrounds this is the conformal deformation detour complex of [1]. Formal self-adjointness holds in all signatures; ellipticity is a Riemannian statement and is not assumed on the Lorentzian cylinder.

2.2 The global four-dimensional deformation complex

The detour sequence (12) is not the only complex available on the conformally flat cylinder. On an oriented four-dimensional conformally flat pseudo-Riemannian manifold, the adjoint BGG deformation complex is [2, 5]

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2 T^*[2]) \xrightarrow{C_1} \Gamma(\mathcal{C}[2]) \xrightarrow{C_1^\sharp \star} \Gamma(S_0^2 T^*[-2]) \xrightarrow{K_0^\sharp} \Gamma(T^*[-2]). \quad (13)$$

Here $\mathcal{C}[2]$ is the algebraic Weyl-curvature bundle and $C_1^\sharp \star$ means $C_1^\sharp \circ \star$, with the Hodge star acting on one antisymmetric index pair. In particular,

$$\boxed{C_1^\sharp \star C_1 = 0.} \quad (14)$$

The repository certificate verifies this identity directly as a constant-coefficient differential-operator identity in both Euclidean and Lorentzian conventions; Gover and Peterson supply the invariant global sequence.

On a locally conformally flat manifold, the flat BGG complex is a fine resolution of the sheaf of parallel sections of the flat adjoint-tractor bundle [6, 5]. The cylinder

$$M = \mathbb{R} \times S^3 \simeq S^3 \quad (15)$$

is simply connected, so that flat local system is the constant system with fibre $\mathfrak{so}(4, 2)_{\mathbb{C}}$. Fine-resolution cohomology therefore gives an all-level global statement without an elliptic argument.

Theorem 2.2 (Global smooth deformation cohomology). *For smooth complexified global sections on the conformal cylinder,*

$$H_{\text{def}}^q(M) \cong H^q(S^3; \mathbb{C}) \otimes \mathfrak{so}(4, 2)_{\mathbb{C}} \cong \begin{cases} \mathfrak{so}(4, 2)_{\mathbb{C}}, & q = 0, 3, \\ 0, & q = 1, 2, 4. \end{cases} \quad (16)$$

In particular, the metric and curvature slots are globally exact:

$$\boxed{\ker C_1 = \text{im } K_0,} \quad \boxed{\ker(C_1^\sharp \star) = \text{im } C_1.} \quad (17)$$

Proof. The fine resolution computes sheaf cohomology of the flat adjoint local system. Simple connectivity trivializes its monodromy, while $\mathbb{R} \times S^3$ deformation retracts onto S^3 . The only nonzero ordinary cohomology groups are in degrees zero and three. Vanishing in degrees one and two is exactly the pair of statements (17). $\square$

The theorem is signature-independent and concerns unrestricted smooth global sections. It does not by itself establish continuity, closed range, or bounded inverses on a Hilbert or Krein completion.

2.3 The Bach quotient as an on-shell curvature space

The passage from deformation-complex exactness to a detour equation is a general exact-sequence argument.

Proposition 2.3 (Detour–curvature lemma). *Let*

$$E_0 \xrightarrow{K} E_1 \xrightarrow{C} E_2 \xrightarrow{D_2} E_3 \quad (18)$$

be a complex which is exact at E_1 and E_2 , and let $G : E_2 \rightarrow E'_1$ be an operator such that the detour equation is $B = GC$. Then $[h] \mapsto Ch$ induces an isomorphism

$$\boxed{\frac{\ker B}{\operatorname{im} K} \cong \ker D_2 \cap \ker G.} \quad (19)$$

Proof. If $Bh = 0$, then $GC h = 0$, while $D_2 Ch = 0$ because the displayed sequence is a complex. If $Ch = 0$, exactness at E_1 gives $h = K\xi$, so the induced map is injective. Conversely, if $U \in \ker D_2 \cap \ker G$, exactness at E_2 gives $U = Ch$; then $Bh = GU = 0$, proving surjectivity. $\square$

The middle local cohomology relevant to the free field is

$$\mathcal{H}_B := \frac{\ker B_{\operatorname{lin}}}{\operatorname{im} K_0}. \quad (20)$$

In the four-dimensional deformation complex,

$$D_2 = C_1^\sharp \star, \quad G = C_1^\sharp, \quad B_{\operatorname{lin}} = GC_1. \quad (21)$$

Define the geometrically on-shell curvature space

$$\mathcal{W}_{\operatorname{geom}} := \ker(C_1^\sharp \star) \cap \ker C_1^\sharp. \quad (22)$$

The smooth global BGG theorem and Proposition 2.3 identify the metric quotient with this space exactly.

Theorem 2.4 (Cylinder detour–curvature isomorphism). *For smooth complexified fields on the conformal cylinder, the curvature map $[h] \mapsto C_1 h$ induces an $SO(4, 2)$ -equivariant isomorphism*

$$\boxed{\frac{\ker B_{\operatorname{lin}}}{\operatorname{im} K_0} \cong \mathcal{W}_{\operatorname{geom}} = \ker C_1^\sharp \cap \ker(C_1^\sharp \star).} \quad (23)$$

Proof. If $B_{\text{lin}}h = 0$ and $U = C_1h$, then the factorization (9) gives $C_1^\sharp U = 0$, while (14) gives $C_1^\sharp \star U = 0$. Thus the curvature map lands in the displayed intersection.

If $C_1h = 0$, exactness at the metric slot in (17) gives $h = K_0\xi$, proving injectivity. Conversely, suppose U lies in both kernels. Exactness at the curvature slot gives $U = C_1h$ for a smooth global h , and then $B_{\text{lin}}h = C_1^\sharp U = 0$. Hence every element of the intersection has a Bach-flat metric potential, proving surjectivity. $\square$

In Lorentz signature, complexified Weyl tensors split into Hodge eigenspaces $U = U_+ + U_-$ with $\star U_\pm = \pm iU_\pm$. If $a = C_1^\sharp U_+$ and $b = C_1^\sharp U_-$, the two target equations are

$$a + b = 0, \quad ia - ib = 0, \quad (24)$$

so $a = b = 0$. Define the smooth on-shell chiral geometric spaces by

$$\mathcal{W}_{\text{geom},\pm}^{\text{sm}} := \{U_\pm : \star U_\pm = \pm iU_\pm, C_1^\sharp U_\pm = 0\}. \quad (25)$$

Then Theorem 2.4 becomes

$$\boxed{\frac{\ker B_{\text{lin}}}{\text{im } K_0} \cong \mathcal{W}_{\text{geom},+}^{\text{sm}} \oplus \mathcal{W}_{\text{geom},-}^{\text{sm}}.} \quad (26)$$

All maps commute with $SO(4,2)$. Hence the quotient-to-curvature map preserves compact energy and both rotation spins on quotient classes. It remains to match this geometric system to the abstract positive-energy module constructed in Section 3.

Theorem 2.5 (Algebraic cylinder realization). *Let $\mathcal{W}_{\text{geom},\pm}^{\text{alg},+}$ denote the D -finite, $SO(4)$ -finite positive-energy solutions of (25). The curvature construction restricts to an isomorphism*

$$\boxed{\mathcal{W}_{\text{geom},+}^{\text{alg},+} \oplus \mathcal{W}_{\text{geom},-}^{\text{alg},+} \cong \mathcal{W}_+ \oplus \mathcal{W}_-.} \quad (27)$$

Every curvature basis vector has an explicit Bach-flat metric representative in the same D -energy and $SO(4)$ type.

Proof. Put $r = \tan(\beta/2)$, $z = 1 + r^2$, and $q = d\beta + i \sin \beta d\gamma$. For positive chirality the normalized highest-weight metric representatives are the radiation-gauge cylinder modes

tower	J	nonzero metric component
E_n	$n/2$	$h_{ij} = N_E(J)T[J]_{ij}e^{-int}$
A_n	$(n-1)/2$	$h_{0i} = N_A(J)V[J]_i e^{-int}$
L_n	$(n-2)/2$	$h_{ij} = N_L(J)T[J]_{ij}e^{-int}$

(28)

Here

$$\begin{aligned} V[J] &= -\frac{\sqrt{2J}}{4\pi} \cos^{2J-1} \frac{\beta}{2} e^{-i[(J+\frac{1}{2})\alpha + (J-\frac{1}{2})\gamma]} q, \\ T[J] &= \frac{\sqrt{2(2J-1)}}{16\pi} \cos^{2J-2} \frac{\beta}{2} e^{-i[(J+1)\alpha + (J-1)\gamma]} q \otimes q, \end{aligned} \quad (29)$$

and

$$N_E = \frac{1}{4\sqrt{J(2J+1)}}, \quad N_A = \frac{1}{2\sqrt{(2J-1)(2J+1)(2J+3)}}, \quad N_L = \frac{1}{4\sqrt{(J+1)(2J+1)}}. \quad (30)$$

An exact coordinate calculation retains symbolic integer n , constructs all components of $C_1 h$, and applies $C_1^\sharp$ independently. After suppressing the common factor $e^{-i(nt+m_L\alpha+m_R\gamma)}z^{-a}$, nonzero pivots are

$$\begin{aligned}(C_1 h_E)_{0202} &= \frac{\sqrt{n(n-1)(n+1)}}{16\pi z^2}, \\(C_1 h_A)_{0102} &= \frac{\sqrt{n(n-2)(n-1)}}{32\pi\sqrt{n+2}z}, \\(C_1 h_L)_{0202} &= \frac{\sqrt{n(n-3)(n-1)}}{16\pi z^2}.\end{aligned}\tag{31}$$

They are nonzero at every allowed tower energy. The full tensors are algebraically trace-free, have one common Hodge chirality, and obey $C_1^\sharp C_1 h = 0$ identically in n . Naturality of C_1 extends each nonzero highest-weight intertwiner to its complete $SO(4)$ irrep. Defining $U_{F,n,M} = C_1 h_{F,n,M}$ therefore gives the explicit same-block inverse

$$R_n U_{F,n,M} = h_{F,n,M}, \quad C_1 R_n = \text{id}, \quad F \in \{E, A, L\}.\tag{32}$$

The orientation-reversing cylinder isometry $\alpha \leftrightarrow \gamma$ exchanges the two $SU(2)$ factors and supplies the other chirality.

For exhaustion, the flat positive-chirality BGG data consist of the lowest-weight curvature primary $(2; 2, 0)$, the equation module $(4; 1, 1)$, and its identity $(5; \frac{1}{2}, \frac{1}{2})$. Their alternating character is

$$\chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}.\tag{33}$$

The cylinder module $\mathcal{W}_+$ is cyclic from the same primary and has this character by (39); parity supplies the other chirality. The nonzero graded intertwiner and equality of every graded $SO(4)$ multiplicity prove exhaustion. Equations (28)–(32) supply finite harmonic metric representatives directly, excluding secular or infinite-harmonic preimages in this algebraic category. The exact coordinate tensor calculation, a machine-readable certificate, and the generated pivot table are produced by `symbolic/verify_conformal_cylinder_preimages.py`. $\square$

2.4 The finite-jet calculation as an independent convention audit

The earlier finite-jet calculation remains useful, but it is no longer a substitute for an unknown all-level theorem. On homogeneous Euclidean polynomial jets of degree $n = 2, \dots, 6$, the matrices for K , C_1 , and B_1 obey

$$C_1 K = 0, \quad B_1 K = 0, \quad \ker C_1 = \text{im } K.\tag{34}$$

The exact ranks are shown in Table 1.

The quotient dimensions $(10, 40, 82, 136, 202)$ agree with the first five levels of the parity-complete $E/A/L$ module introduced below. The same calculation separates a 15-dimensional kernel of the low-degree gauge map, generated by four translations, six rotations, one dilation, and four special conformal transformations. These finite results are independent regression rails for (17) and (27) in the repository normalization.

n	$\dim G_n$	$\dim h_n$	$\dim W_{n-2}$	$\dim B_{n-4}$	$\text{rank } K$	$\text{rank } C_1$	$\text{rank } B_1$	$\dim(\ker B_1 / \text{im } K)$
2	90	100	10	0	90	10	0	10
3	160	200	40	0	160	40	0	40
4	259	350	100	10	259	91	9	82
5	392	560	200	40	392	168	32	136
6	564	840	350	100	564	276	74	202

Table 1: Exact homogeneous-polynomial detour ranks. The finite Euclidean jet calculation independently audits the conventions and the first five levels of the global BGG theorem.

2.5 The finite global zero-mode pair

The same topology that kills the metric and curvature cohomology leaves two finite-dimensional groups:

$$H_{\text{def}}^0(M) \cong \mathfrak{so}(4, 2)_{\mathbb{C}}, \quad H_{\text{def}}^3(M) \cong \mathfrak{so}(4, 2)_{\mathbb{C}}. \quad (35)$$

The degree-zero copy is the familiar space of fifteen conformal Killing fields. The degree-three copy lies on the adjoint equation side of the self-dual deformation complex. Refining both groups by compact weight gives

$$4_{-1} \oplus 7_0 \oplus 4_{+1}. \quad (36)$$

Poincaré duality on S^3 and the Killing form pair the two copies, with

$$\kappa(\mathfrak{g}_r, \mathfrak{g}_s) = 0 \quad \text{unless} \quad r + s = 0. \quad (37)$$

Its dimension, representation, and position are exactly those expected of the dual residual constraint sector. It is therefore a strong candidate for the geometric source of the fifteen moment-map or Taub components constructed in Section 4. That identification is not yet a theorem: the full four-dimensional BGG sequence must first be kept distinct from the shorter Bach detour complex, and the degree-three bundle representative must be located explicitly. Importantly, the direct second-order argument in Proposition 4.1 establishes the Taub constraints without depending on this identification. If the BGG and BFV realizations are then compared, equivariance reduces the map to one nonzero component and one overall scalar.

Thus the smooth deformation complex contains a controlled finite pair, not an unspecified collection of extra global classes. A complete BV argument must still place the degree-three copy at the correct ghost/antifield degree and construct the corresponding zero-mode projector.

2.6 Domain boundary of the theorem

Theorems 2.2 and 2.4 are exact for smooth complexified global sections. Theorem 2.5 additionally identifies the D -finite, $SO(4)$ -finite positive-energy submodule by explicit metric representatives. These theorems do not identify spacelike-compact or distributional cohomology, nor do they prove continuity and closed range on an infinite Hilbert/Krein completion. If such an analytic completion is required, Green-hyperbolic complexes and causal homotopies remain natural tools [4].

The other remaining issue is BV rather than local geometry. The fifteen conformal-Killing zero modes and their degree-three duals must be split from the local complex without double counting. Dneprov and Grigoriev's minimal nonlinear Weyl-jet model supplies compatible presymplectic data for this comparison [19], but its curvature-dependent ghost terms must not

be imported as a proof that the free residual transfer is strict. It does not perform the global cylinder BFV split.

3 The on-shell Weyl module

3.1 Character and cylinder towers

We now define the coefficient module used in the exact residual calculation. A positive-chirality Weyl curvature is a conformal primary with labels

$$(\Delta; j_L, j_R) = (2; 2, 0), \quad (38)$$

and the opposite chirality is its parity conjugate. The Bach equation is in $(4; 1, 1)$ and its differential identity is in $(5; \frac{1}{2}, \frac{1}{2})$. The resulting one-chirality character is

$$\chi_+(q) = \chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \quad (39)$$

This defines the abstract algebraic positive-energy module $\mathcal{W}_+$. The character and tower calculation below, together with the explicit intertwiner and metric preimages of Theorem 2.5, identify it with the geometric chiral system.

On $\mathbb{R} \times S^3$, $\mathcal{W}_+$ decomposes into three multiplicity-free towers:

$$E_n^+ : \left(\frac{n+2}{2}, \frac{n-2}{2} \right), \quad \dim E_n = (n+3)(n-1), \quad n \geq 2, \quad (40)$$

$$A_n^+ : \left(\frac{n}{2}, \frac{n-2}{2} \right), \quad \dim A_n = (n+1)(n-1), \quad n \geq 3, \quad (41)$$

$$L_n^+ : \left(\frac{n}{2}, \frac{n-4}{2} \right), \quad \dim L_n = (n+1)(n-3), \quad n \geq 4. \quad (42)$$

The negative-chirality towers exchange j_L and j_R . Summing (40)–(42) reproduces (39). For both chiralities the first levels have dimensions

$$\dim \mathcal{H}_2 = 10, \quad \dim \mathcal{H}_3 = 40, \quad \dim \mathcal{H}_4 = 82, \quad \dim \mathcal{H}_5 = 136, \quad \dim \mathcal{H}_6 = 202. \quad (43)$$

The labels E, A, L are cylinder oscillator labels only. In particular, they should not be interpreted as three independently signable conformal representations. Proper conformal transformations mix the towers.

3.2 Invariant form and exact generator action

In normalized cylinder coordinates the unique standard-real, parity-compatible invariant form is, up to overall convention,

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \quad (44)$$

It is nondegenerate and indefinite. The proper conformal lowering and raising generators satisfy

$$(K_q^-)^\sharp = K_q^+, \quad X^\sharp = X \quad (X = D, SU(2)_L, SU(2)_R), \quad (45)$$

where $X^\sharp = J_{\text{conf}}^{-1} X^\dagger J_{\text{conf}}$.

Multiplicity one reduces each allowed lowering block to a scalar reduced coefficient. With source energy n , the six stable families are

$$\begin{aligned}
c_{EE}(n) &= \sqrt{\frac{2(n-1)(n+1)(n+3)}{n+2}}, & c_{AE}(n) &= \sqrt{\frac{8(n-1)}{(n-2)(n+2)}}, \\
c_{AA}(n) &= \sqrt{\frac{2(n-3)(n-1)(n+2)}{n-2}}, & c_{LE}(n) &= -\sqrt{\frac{2(n-3)}{n-2}}, \\
c_{LA}(n) &= \sqrt{\frac{8}{n-2}}, & c_{LL}(n) &= \sqrt{2(n-2)(n+1)}.
\end{aligned} \tag{46}$$

Together with Clebsch–Gordan coefficients and (45), these determine the all-level one-particle $\mathfrak{so}(4, 2)$ action. The exact reconstruction verifies the conformal commutators on every interior level of a finite energy buffer; the top shell is used only as a buffer and is never mistaken for a finite conformal representation.

The free Fock coefficient module is the bosonic second quantization

$$\mathcal{F}_{\mathcal{W}} = \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \tag{47}$$

with multiplicative lift of (44). The residual generators preserve oscillator particle number.

4 Hamiltonian moment-map normalization

The residual action is not introduced as an arbitrary matrix representation. In oscillator coordinates z , write the symplectic form as

$$\Omega = i d\bar{z} J \wedge dz. \tag{48}$$

If a conformal generator X acts linearly by $\delta_X z = K_X z$, its quadratic Hamiltonian moment map is

$$\mu_X(z, \bar{z}) = \bar{z} M_X z, \quad M_X = J K_X, \quad d\mu_X = \iota_{X_X} \Omega. \tag{49}$$

The J -adjoint identities ensure the correct reality of μ_X .

4.1 The conformal Taub constraint

The same residual parameters arise as reducibilities of the local $\text{Diff} \times \text{Weyl}$ complex. Let

$$\epsilon = (\xi, \sigma), \quad K(\xi, \sigma) = 0, \quad \sigma = -\frac{1}{4} \bar{\nabla}_\mu \xi^\mu, \tag{50}$$

and let h satisfy $B^{(1)}[h] = 0$. With the Taylor convention

$$g(\lambda) = \bar{g} + \lambda h + \frac{\lambda^2}{2} k + O(\lambda^3), \tag{51}$$

the second-order equation is

$$B^{(1)}[k] + B^{(2)}[h, h] = 0. \tag{52}$$

Proposition 4.1 (Conformal Taub solvability constraint). *For every residual conformal reducibility ϵ , the current*

$$J_\epsilon^\mu[h] := \xi_\nu B^{(2)\mu\nu}[h, h] \quad (53)$$

is conserved. If h is tangent to a second-order family of Bach-flat metrics on the compact cylinder, then

$$\boxed{\mathcal{T}_\epsilon[h] := \int_{S^3} n_\mu \xi_\nu B^{(2)\mu\nu}[h, h], d\Sigma = 0.} \quad (54)$$

Proof. At second order around a Bach-flat background, the exact divergence and trace identities imply $\bar{\nabla}_\mu B^{(2)\mu\nu}[h, h] = 0$ and $B^{(2)\mu}{}_\mu[h, h] = 0$ whenever $B^{(1)}[h] = 0$. Hence

$$\bar{\nabla}_\mu J_\epsilon^\mu = B^{(2)\mu\nu} \bar{\nabla}_{(\mu} \xi_{\nu)} = -\sigma B^{(2)\mu}{}_\mu = 0. \quad (55)$$

If (52) has a solution, the charge equals minus the integral of $\xi_\nu B^{(1)\mu\nu}[k]$. For a reducibility parameter, the linearized Noether current is a spatial divergence. Since $\partial S^3 = \emptyset$, its integral vanishes. This is the standard compact-Cauchy linearization-stability argument [21, 22]. $\square$

The covariant phase-space Noether identity predicts that this same quadratic constraint is the Hamiltonian of the residual linear action:

$$\mu_\epsilon(h) = \frac{1}{2} \Omega_\Sigma(h, R_\epsilon h) \stackrel{?}{=} \pm \mathcal{T}_\epsilon[h]. \quad (56)$$

The sign and a possible factor of two depend only on the Taylor convention for $B^{(2)}$. Both sides form equivariant adjoint/coadjoint $\mathfrak{so}(4, 2)$ modules. Since the complexified adjoint module is simple, an equivariant nonzero comparison is unique up to one scalar. Consequently the all-component equality can be proved by establishing equivariance and matching one generator, preferably D . This is also the local-BRST relation between reducibilities and conserved-current classes [16, 15].

In the curved-cylinder wave normalization used for the independent Taub calculation, comparison with the quadratic action fixes the proper-conformal kernel to

$$M_{\text{Taub}} = -\frac{\sqrt{2}}{4\pi} JK^- \quad (57)$$

up to the explicitly declared orientation and harmonic phases. Two independent curvature components fix the common coefficient; it is not a one-entry fit. An exact two-chirality jet through source energy four verifies all fifteen moment-map components and the full conformal algebra on interior energies two and three.

Equation (57) verifies (56) for the tested proper-conformal kernels. The remaining all-level task is the single equivariant normalization comparison, not fifteen unrelated curvature calculations.

This section supplies the Hamiltonian normalization of the residual action and its Taub interpretation. The cohomology theorem below depends only on the exact module action, not on interpreting a finite moment-map jet as a physical quotient.

5 Residual BRST and Cartan homotopy reduction

5.1 The absolute residual complex

Let

$$\mathfrak{g} = \mathfrak{so}(4, 2) \quad (58)$$

with compact grading

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \quad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4). \quad (59)$$

Here D and $\mathfrak{so}(4)$ lie in $\mathfrak{g}_0$, while proper conformal lowering and raising generators lie in $\mathfrak{g}_{-1}$ and $\mathfrak{g}_{+1}$. The dual ghosts carry the opposite compact degree.

For a coefficient module $V \subset \mathcal{F}_{\mathcal{W}}$, the absolute residual complex is

$$C^q(V) = V \otimes \Lambda^q \mathfrak{g}^*, \quad (60)$$

with Chevalley–Eilenberg differential

$$d = c^a \rho(G_a) - \frac{1}{2} f^a_{bc} c^b c^c \iota_a. \quad (61)$$

It raises ghost number by one and preserves the total compact degree

$$\delta = D_{\text{matter}} + D_{\text{ghost}}. \quad (62)$$

The standard centered polarization uses the product v of the four ghosts dual to the raising generators. It has

$$\text{gh}(v) = 4, \quad D_{\text{ghost}}(v) = -4. \quad (63)$$

This is the residual ghost convention used in the cylinder construction of [11]. Its compact degree and uniqueness as the bottom rotational scalar follow from the $(4, 7, 4)$ exterior-algebra polarization, and Lemma 6.1 shows that its complementary-degree pairing is canonical up to one overall volume. Only the match of that scalar to the strict-pure-Weyl BV/BFV normalization remains a hypothesis in Section 8; the broader Riegert sector of [11] is not included.

5.2 Full residual gauging is a physical assumption

Hamada’s background-independence programme identifies the transformations which preserve the cylinder gauge with residual diffeomorphisms and imposes the resulting conformal charges through BRST [10, 11]. This supplies a direct precedent for the complex (60). It does not prove that every quantization of strict pure-Weyl gravity must use the same quotient.

There are at least two distinct cylinder problems.

1. In the *fully gauged residual problem*, the cylinder is a gauge representative of a closed generally covariant system. All fifteen conformal-Killing transformations, including compact time translation, are generated by residual constraints. The ghost for D exists and the Cartan homotopy below is an allowed operation.
2. In the *fixed-background cylinder problem*, D is the Hamiltonian and $SO(4, 2)$ is a global symmetry acting on states or observables. Its eigenspaces are not quotiented. The D ghost and its contraction cannot be used to erase nonzero energy.

Proposition 4.1 gives the first choice a direct linearization-stability interpretation. Within the closed-universe Dirac boundary problem, the slice S^3 has no spatial boundary supporting an independent asymptotic charge, and the residual conformal Killing parameters instead generate quadratic integrability constraints, including the D constraint. Subject to the all-level normalization in (56), full residual gauging is therefore a coherent zero-boundary-charge sector of pure conformal gravity; it is not a device introduced merely to erase positive cylinder energy. Compactness and the Taub identity support this interpretation but do not make it the unique physical quantization.

Adding a boundary or asymptotic region, coupling an external clock, or deparametrizing the cylinder can convert D into a physical global charge. That is the second problem above, not a contradiction of the first. Because $[K^-, K^+]$ contains D , the second problem is not obtained by simply deleting one ghost while leaving the rest of (61) unchanged. It requires a different relative or boundary-charge complex. Theorem 6.2 answers only the first problem. A strict pure-Weyl derivation must state the temporal boundary conditions, the residual gauge group, and the treatment of its charges before calling the result a physical state space.

5.3 Cartan homotopy

Let ι_D be contraction with the generator D in the residual ghost exterior algebra. The usual Cartan identity is

$$d\iota_D + \iota_D d = \mathcal{L}_D. \quad (64)$$

Theorem 5.1 (Cartan homotopy reduction to compact degree zero). *Decompose the residual complex into total compact-degree eigenspaces,*

$$C = \bigoplus_{\delta} C_{\delta}, \quad \mathcal{L}_D|_{C_{\delta}} = \delta \text{ id}. \quad (65)$$

Every C_{δ} with $\delta \neq 0$ is contractible. Hence inclusion of the centered subcomplex is a quasi-isomorphism:

$$\boxed{H^{\bullet}(C, d) \cong H^{\bullet}(C_{\delta=0}, d).} \quad (66)$$

Proof. On C_{δ} with $\delta \neq 0$, define

$$h_{\delta} = \frac{\iota_D}{\delta}. \quad (67)$$

Equation (64) gives

$$dh_{\delta} + h_{\delta}d = \text{id}. \quad (68)$$

Thus every closed vector of nonzero total degree has the explicit primitive $\delta^{-1}\iota_D\Psi$. If $\mathcal{L}_D$ has a nilpotent part, the same proof works wherever $\mathcal{L}_D$ is invertible by using $h = \iota_D\mathcal{L}_D^{-1}$. $\square$

The theorem depends on treating D as a residual gauge generator. If cylinder time translation is retained as a physical Hamiltonian or a boundary charge, contraction by its ghost is not an allowed quotient and (66) does not apply.

5.4 Finite centered inventory

At ghost number four, the minimal residual exterior algebra contains at most the four negative-degree ghosts, so

$$D_{\text{ghost}} \geq -4. \quad (69)$$

At $\delta = 0$ this implies

$$D_{\text{matter}} \leq 4. \quad (70)$$

Every nonvacuum Weyl quantum has weight at least two. Therefore only particle numbers $N = 0, 1, 2$ can contribute to the centered ghost degree selected by the Hamada polarization. All $N \geq 3$ sectors have matter weight at least six and are contractible before any matrix-rank computation.

6 Complete centered cohomology

6.1 The relative weight-four kernel

The complete matter Fock space at weight four is

$$\mathcal{F}_4 = \mathcal{H}_4 \oplus \text{Sym}^2 \mathcal{H}_2, \quad \dim \mathcal{F}_4 = 82 + 55 = 137. \quad (71)$$

The induced matter form has signature

$$\text{sig}(J_{\mathcal{F}_4}) = (97, 40). \quad (72)$$

The relative conformal-primary conditions are

$$(D - 4)|\Psi\rangle = 0, \quad R|\Psi\rangle = 0, \quad K_q^-|\Psi\rangle = 0. \quad (73)$$

Their exact kernel on the full space (71) is two-dimensional. In a spin-two magnetic basis its normalized chiral generators are

$$|W_{\pm}^2\rangle = \sqrt{\frac{2}{5}}|2, -2\rangle_{\pm} - \sqrt{\frac{2}{5}}|1, -1\rangle_{\pm} + \frac{1}{\sqrt{5}}|0, 0\rangle_{\pm}. \quad (74)$$

They obey

$$\langle W_r^2 | J_{\mathcal{F}} | W_s^2 \rangle = \delta_{rs}. \quad (75)$$

This relative computation is not by itself an absolute cohomology theorem; incoming exact states must also be excluded.

There is a simple representation-theoretic reason for the number two. The weight-two one-particle space is

$$E_2^+ \oplus E_2^- = (2, 0) \oplus (0, 2). \quad (76)$$

The symmetric square decomposes as

$$\text{Sym}^2(2, 0) \oplus [(2, 0) \otimes (0, 2)] \oplus \text{Sym}^2(0, 2). \quad (77)$$

There is one compact scalar in each same-chirality symmetric square and no scalar in the mixed product. Since $E_2^{\pm}$ are lowest-weight spaces, every K_q^- annihilates both factors, so these two scalar singlets are relative primaries. By contrast, the one-particle weight-four space contains

$$(3, 1) \oplus (2, 1) \oplus (2, 0) \quad (78)$$

and its parity conjugate, with no compact scalar. Thus representation theory predicts the two relative candidates before the matrix calculation. The absolute computation below is still necessary to exclude exactness and to prove the one-particle acyclicity. A complete Hochschild–Serre or Kostant-style derivation of the latter ranks would be a useful conceptual replacement for the large matrices [5].

6.2 Absolute kernel and image

The cutoff-complete absolute calculation supplies that missing test. For one chirality in the one-particle coefficient module, the centered complex around ghost number four has exact dimensions

$$C_0^3(290) \xrightarrow{d_3} C_0^4(1311) \xrightarrow{d_4} C_0^5(3657). \quad (79)$$

The entries lie in $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5})$. Good-prime reduction modulo 241, combined with exact nilpotency, fixes the characteristic-zero ranks to

$$\text{rank } d_3 = 260, \quad \text{rank } d_4 = 1051. \quad (80)$$

Since the ranks sum to 1311,

$$H_{\delta=0, N=1}^4 = 0. \quad (81)$$

Parity supplies the identical result for the opposite chirality.

The lowest two-particle block begins at matter weight four. Exterior saturation makes its beginning

$$0 \longrightarrow C_0^4(55) \xrightarrow{d_4} C_0^5(385) \xrightarrow{d_5} C_0^6(1155). \quad (82)$$

There is no incoming C_0^3 space: three residual ghosts have compact degree at least -3 and cannot center a two-particle coefficient of minimum weight four. Exact nilpotency gives $d_5 d_4 = 0$. The two vectors (74) lie in $\ker d_4$, while a good-prime minor gives $\text{rank } d_4 \geq 53$. The displayed independent kernel vectors give $\text{rank } d_4 \leq 53$. Therefore

$$H_{\delta=0, N=2}^4 = \text{span}\{[W_+^2], [W_-^2]\}. \quad (83)$$

For the vacuum coefficient module the complex is ordinary Lie-algebra cohomology. Since

$$H^\bullet(\mathfrak{so}(4, 2); \mathbb{C}) \cong \Lambda(u_3, u_5, u_7), \quad (84)$$

one has

$$H_{\delta=0, N=0}^4 = 0. \quad (85)$$

6.3 The canonical residual CE pairing

The residual ghost normalization is fixed almost completely by the compact grading. Write

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \quad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4), \quad (86)$$

and choose nonzero determinant vectors

$$v_- \in \det(\mathfrak{g}_{+1}^*), \quad \Theta_0 \in \det(\mathfrak{g}_0^*), \quad v_+ \in \det(\mathfrak{g}_{-1}^*), \quad (87)$$

with $v_+ = v_-^\dagger$. Their product

$$\Omega_{\mathfrak{g}} = v_+ \wedge \Theta_0 \wedge v_- \quad (88)$$

is the top Berezin volume on all fifteen residual ghosts. Define the polarized pairing by coefficient extraction,

$$(\alpha, \beta)_{\text{gh}} := [\alpha^\dagger \wedge \Theta_0 \wedge \beta]_{\Omega_{\mathfrak{g}}}. \quad (89)$$

Lemma 6.1 (Canonical centered ghost pairing). *After fixing the orientation and the overall residual volume, (89) is the unique determinant-saturating pairing compatible with the $(4, 7, 4)$ polarization. It obeys*

$$(v_-, v_-)_{\text{gh}} = 1. \quad (90)$$

Because $\mathfrak{so}(4, 2)$ is semisimple and hence unimodular, the Chevalley–Eilenberg differential is graded skew for the associated complementary-degree Poincaré pairing.

Proof. Each of the three determinant lines is one-dimensional. Once their product is oriented and normalized, coefficient extraction gives (90) and leaves no matrix freedom. Unimodularity makes the top CE volume invariant, so integration by parts in the exterior algebra gives graded skew-adjointness of the CE differential. $\square$

Consequently exact cochains are orthogonal to closed complementary-degree cochains, and the pairing descends to residual CE cohomology.

In Hamada’s oscillator notation, v_- is the product of the four proper-conformal ghosts, Θ_0 contains the dilation/time ghost and the six rotation ghosts, and the bra supplies the four conjugate ghosts [11]. The factor of i is a real-structure convention. The repository exterior-algebra certificate verifies the saturation, Hermiticity, compact degree -4 , and unit overlap directly.

The fixed insertion is not a nondegenerate inner product on every individual ghost-number block. It is the polarized form of the canonical complementary-degree CE pairing. It is sufficient here because the incoming two-particle space is empty and the two normalized representatives are fixed. Multiplying (90) by the matter restriction (75) gives their exact residual cohomological Gram matrix. It is Hermitian and sesquilinear on the complex oscillator module; the parity-compatible real structure gives its real restriction. This is an intrinsic statement about the chosen residual CE complex. We reserve the phrase “transferred pure-Weyl BV cohomological pairing” for the cyclic-transfer statement in Section 8. Showing that the full local BV/BFV reduction induces this residual orientation and overall volume is part of the conditional bridge.

Theorem 6.2 (Complete cohomology under full residual conformal gauging). *Let the coefficient module be the free bosonic Fock space $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$, let the residual algebra be $\mathfrak{so}(4, 2)$, treat D as a gauge generator, and use the centered ghost polarization (63). Then the complete minimal absolute residual cohomology at the selected ghost number four is*

$$\boxed{H_{\text{res}, \min}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.} \quad (91)$$

There is no one-particle class and no higher-matter-weight class in this minimal centered polarization.

Proof. The Cartan homotopy, Theorem 5.1, eliminates every nonzero total compact degree. The ghost-degree floor then reduces the centered inventory to $N = 0, 1, 2$. Equations (85), (81), and (83) exhaust those sectors. Equations (75) and (90) give the pairing. $\square$

Remark 6.3 (What is positive). The theorem does not turn the indefinite one-particle module (44) into a positive particle space. It removes the one-particle sector from this absolute residual cohomology. The positive I_2 belongs to two ghost-dressed scalar vertex classes.

7 Descent and the dynamical/topological split

7.1 Why weight four is selected

Let V_h be a scalar conformal primary of weight h , and let

$$\omega = \frac{1}{4!} \epsilon_{\mu\nu\rho\sigma} c^\mu c^\nu c^\rho c^\sigma \quad (92)$$

be the top lowering-ghost product. In the residual descent conventions,

$$sV_h = c^\mu \nabla_\mu V_h + \frac{h}{4} (\nabla_\mu c^\mu) V_h, \quad s\omega = -\omega \nabla_\mu c^\mu, \quad \omega c^\mu = 0. \quad (93)$$

It follows that both the integrated density and the unintegrated ghost-dressed operator have the same closure defect, proportional to $h/4 - 1$. Hence

$$\boxed{[\omega V_4] \longleftrightarrow \left[\int_M d^4x \sqrt{-g} V_4 \right]}. \quad (94)$$

The -4 compact degree of the ghost factor is therefore the conformal counterpart of the four-dimensional volume form, rather than an accidental normal-ordering shift. This weight-four state-operator mechanism and a Weyl-square example were already explicit in [11]; the new use here is the absolute exhaustion and pairing calculation of Section 6.

Three cohomological quotients appear and must not be identified without the transfer theorem:

$$H^4(\mathfrak{so}(4, 2); \mathcal{F}_W), \quad H_{\text{loc}}^{0,4}(s \mid d), \quad \frac{\{\text{integrated local functionals}\}}{\{\text{allowed boundary functionals}\}}. \quad (95)$$

The first is the residual Chevalley–Eilenberg group computed exactly in this paper. The second is local BRST cohomology modulo spacetime descent [15]. The third depends on topology and boundary conditions. Equation (94) identifies representatives in the residual construction; its promotion to an isomorphism with the strict pure-Weyl local group is part of the conditional bridge.

7.2 Parity basis

Parity exchanges W_+^2 and W_-^2 . Define

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \quad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}. \quad (96)$$

Up to the orientation-dependent factor of i in Lorentzian self-duality conventions, descent identifies

$$e \sim C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}, \quad o \sim C_{\mu\nu\rho\sigma} \tilde{C}^{\mu\nu\rho\sigma}. \quad (97)$$

Because (96) is orthogonal, its Gram matrix remains I_2 .

7.3 Euler–Lagrange quotient

Let

$$\mathcal{E} : H_{\text{res,min}}^4 \longrightarrow \{\text{local equation and gauge deformations}\} \quad (98)$$

be the Euler–Lagrange map obtained by varying the integrated density. In the conventions of Section 2,

$$\mathcal{E}(e) = \lambda_B B_{\mu\nu}, \quad \lambda_B \neq 0, \quad \mathcal{E}(o) = 0. \quad (99)$$

The first identity is the Weyl-square variation. Under locality, smoothness, Lorentz invariance, and a four-derivative bound, it is the unique nontrivial self-interaction of a single linear conformal graviton modulo equivalence [14].

The second identity is Chern–Weil transgression. In a local frame,

$$P_4 = \text{Tr}(R \wedge R) = dQ_3(\Gamma), \quad (100)$$

where

$$Q_3(\Gamma) = \text{Tr} \left(\Gamma \wedge d\Gamma + \frac{2}{3} \Gamma \wedge \Gamma \wedge \Gamma \right). \quad (101)$$

Its variation is

$$\delta P_4 = 2d \text{Tr}(\delta\Gamma \wedge R), \quad (102)$$

using the Bianchi identity. On a finite cylinder, in a chosen global frame,

$$\int_{[t_1, t_2] \times S^3} P_4 = \text{CS}[\Gamma(t_2)] - \text{CS}[\Gamma(t_1)]. \quad (103)$$

Thus fixed-endpoint or compactly supported variations have no bulk Euler–Lagrange derivative.

Equations (99)–(103) give the exact sequence

$$0 \longrightarrow \text{span}\{o\} \longrightarrow H_{\text{res,min}}^4 \xrightarrow{\mathcal{E}} \text{span}\{B_{\mu\nu}\} \longrightarrow 0. \quad (104)$$

Quotienting variationally trivial local bulk functionals yields

$$H_{\text{dyn}}^4 = H_{\text{res,min}}^4 / \ker \mathcal{E} \cong \text{span}\{[C^2]\}, \quad \boxed{G_{\text{dyn,res}} = I_1}. \quad (105)$$

Equivalently,

$$\boxed{I_2 = I_1^{\text{dynamical}} \oplus I_1^{\text{topological}}}. \quad (106)$$

The odd class is only locally and variationally trivial. A gravitational theta parameter can affect boundaries, horizons, contact terms, large frame transformations, and nontrivial topology [24]. We do not quotient those global data without specifying the boundary problem.

8 The local-to-residual transfer

Theorem 6.2 begins with the locally reduced Weyl module. This section states exactly what is needed to identify it with the complete free pure-Weyl BV answer.

8.1 The free Fock lift is formal

Proposition 8.1 (Algebraic symmetric-algebra lift). *Let (C, q) be the algebraic one-particle free complex over characteristic zero, with q extended as a derivation to $\text{Sym } C$. Then*

$$\boxed{H(\text{Sym } C, q) \cong \text{Sym } H(C, q).} \quad (107)$$

In particular, once the one-particle coefficient module is $\mathcal{W}_+ \oplus \mathcal{W}_-$, its free many-particle cohomology is $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$.

Proof. Degree by degree, $C^{\otimes N}$ obeys the Künneth theorem. Taking symmetric coinvariants under the finite group S_N is exact in characteristic zero, so $H(\text{Sym}^N C, q) \cong \text{Sym}^N H(C, q)$. Taking the algebraic direct sum over N proves (107). $\square$

Thus the Fock-space coefficient module is not an independent field-theory hypothesis. The statement is algebraic; topological tensor products and completed Fock spaces require their own continuity assumptions.

8.2 Equivariant Cartan transfer

Let (C, q) be the gauge-fixed local free BV complex after the fifteen conformal-Killing zero modes have been separated, and let $H = H(q)$. Suppose there is a strong deformation retract

$$(H, 0) \xrightleftharpoons[p]{j} (C, q), \quad jp = \text{id} - qs - sq, \quad pj = \text{id}_H. \quad (108)$$

Let Δ contain the residual zero-mode differential and mixed terms, so $Q = q + \Delta$. With the sign convention in (108), homological perturbation gives

$$I = (\text{id} + s\Delta)^{-1}j, \quad P = p(\text{id} + \Delta s)^{-1}, \quad Q_H = p\Delta(\text{id} + s\Delta)^{-1}j. \quad (109)$$

If

$$[D, q] = [D, \Delta] = [D, s] = 0, \quad pD = D_H p, \quad Dj = jD_H, \quad (110)$$

then the transferred operators

$$\iota_D^H = P\iota_D I, \quad \mathcal{L}_D^H = P\mathcal{L}_D I \quad (111)$$

obey

$$[Q_H, \iota_D^H]_+ = \mathcal{L}_D^H. \quad (112)$$

Thus every transferred subspace on which $\mathcal{L}_D^H$ is invertible is again contractible.

8.3 Cyclic transfer and the pairing

Let $\sharp$ be the graded adjoint for the full BV/Krein pairing. Suppose the unperturbed retract is cyclic,

$$j^\sharp = p, \quad (s\Delta)^\sharp = \Delta s. \quad (113)$$

Then taking the adjoint of (109) gives

$$I^\sharp = P. \quad (114)$$

Since $PI = \text{id}_H$,

$$\boxed{I^\sharp I = \text{id}_H.} \quad (115)$$

The dressed inclusion is therefore an exact isometry: contractible local dressing changes representatives without changing the reduced Gram matrix.

Proposition 8.2 (Blockwise cyclic retract). *Let a finite compact-energy and $SO(4)$ block carry a nondegenerate graded pairing, let $q^\sharp = -q$, and suppose the induced pairing on $H(q)$ is nondegenerate. Then the block admits a cyclic strong deformation retract. The algebraic direct sum of these retracts may be chosen compact-degree and $SO(4)$ equivariant.*

Proof. Write $B = \text{im } q$ and choose a nondegenerate representative space $\mathcal{H} \subset \ker q$. In $\mathcal{H}^\perp$, choose an isotropic complement L to B . Then $q : L \rightarrow B$ is an isomorphism. Define $s|_B = (q|_L)^{-1}$ and set $s = 0$ on $\mathcal{H} \oplus L$. The decomposition $C = \mathcal{H} \oplus B \oplus L$ gives $qs + sq = 1 - \mathcal{H}p$, while the hyperbolic pairing of B and L gives the cyclic adjoint relation. Choosing the splitting in each finite symmetry block and taking the algebraic direct sum preserves compact degree and $SO(4)$ type. $\square$

This proposition does not provide one splitting which is simultaneously equivariant under the noncompact generators $K^\pm$. Such a splitting cannot be obtained by averaging over the nonunitary conformal module, which need not be semisimple. Compatibility with the full $SO(4, 2)$ action, the global zero-mode split, and the BV pairing remains a field-theory obligation. Boundedness on a completed cylinder space is a separate analytic question.

8.4 Raw polynomial instantiation and measured noncompact defects

The abstract discussion can be instantiated before choosing an analytic completion. In the finite polynomial conformal category, fixed total compact energy gives the exact rational chain

$$G_n \xrightarrow{A_n} M_n \xrightarrow{B_n} E_n \xrightarrow{C_n} I_n, \quad (116)$$

where the four slots contain the Diff $\times$ Weyl ghosts, trace-free metric and trace doublet, Bach equation/metric-antifield row, and Noether identity/ghost-antifield row. The Weyl scalar is written as

$$\omega = \sigma + \frac{1}{4}\partial \cdot \xi, \quad (117)$$

so the vector arrow is the trace-free conformal-Killing operator and the trace is an explicit doublet. The raw matrices commute with all four coordinate translations and special-conformal vector fields, not only with $D \times SO(4)$.

Computational Proposition 8.3 (Raw metric-BV retract and conformal transfer). *In the complete energy buffer $n = 2, 3, 4, 5$, exact rational changes of basis give maps p_n, j_n, s_n satisfying*

$$p_n j_n = 1, \quad j_n p_n = 1 - q_n s_n - s_n q_n, \quad (118)$$

with cohomology dimensions 10, 40, 82, 136. The chosen retract is not strictly equivariant under the noncompact generators. For one Cartesian component, the defect ranks are

	rank ρ_H	rank δ_j	rank δ_p	rank δ_s
P_0	40	29	5	35
K_0	82	80	9	97.

(119)

Every inclusion defect is explicitly q -exact, every projection defect has an explicit right q -homotopy, and the homotopy defect obeys the induced projector identity. The transferred maps obey all sixteen $[K_b, P_a]$ brackets exactly on cohomology. Moreover,

$$p\rho_a s\rho_b j = 0 \quad (120)$$

on the physical metric row for every pair of proper-conformal components; all higher terms vanish because a second s acts on the gauge row.

Thus full equivariance of an arbitrarily chosen representative inclusion is not needed and, for this natural exact splitting, is false. What transfers strictly is the conformal action on cohomology. The calculation therefore realizes the homotopy-equivariant outcome anticipated above rather than silently replacing it by a strict SDR. The exact matrices, defect homotopies, and generated table are produced by `symbolic/verify_conformal_raw_bv_transfer.py`.

8.5 The filtered local-to-residual complex

The possible failure modes can be displayed without compressing them into the phrase “no other row.” After the conformal-Killing zero modes have been split from the local ghosts, introduce the residual-ghost filtration on the bicomplex

$$C^{p,q} = \Lambda^p \mathfrak{g}^* \widehat{\otimes} C_{\text{local}}^q, \quad F^p C = \bigoplus_{r \geq p} C^{r,*}. \quad (121)$$

Here p is residual ghost number and q is local BV degree. A transferred degree-one differential has the filtered expansion

$$Q_H = q_{\text{local}} + d_{\text{res}} + \sum_{k \geq 2} Q_{[k]}, \quad (122)$$

with bidegrees

$$|q_{\text{local}}| = (0, 1), \quad |d_{\text{res}}| = (1, 0), \quad |Q_{[k]}| = (k, 1 - k). \quad (123)$$

The terms $Q_{[k]}$ encode higher transferred ghost operations; assuming that they vanish is a substantive strictness assertion, not automatic homological bookkeeping.

The resulting spectral sequence begins

$$E_1^{p,q} = \Lambda^p \mathfrak{g}^* \otimes H^q(q_{\text{local}}), \quad E_2^{p,q} = H^p(\mathfrak{g}; H^q(q_{\text{local}})), \quad (124)$$

and

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}. \quad (125)$$

For the desired term $E_r^{4,0}$, a higher differential can enter from $(4-r, r-1)$ for $r = 2, 3, 4$, and can leave toward $(4+r, 1-r)$ for $2 \leq r \leq 11$. Thus local classes in degrees $q = 1, 2, 3$ and antifield or negative-degree classes down to $q = -10$ are potential modifiers. They must be inventoried or excluded; it is not enough to compute only the local degree-zero row. Since $0 \leq p \leq 15$, this filtration is bounded. On each finite compact-energy block the cochain spaces used here are finite, so the spectral sequence converges. An infinite completed state space additionally requires a complete, separated filtration. Even after collapse, a cyclic splitting is needed to control extension data and the Hermitian form.

This is the standard local-BRST setting in which antifield and gauge-fixed cohomologies must be compared carefully [15, 17].

Proposition 8.4 (Single-row strictness and collapse). *Suppose the global zero modes have been split exactly, $H^q(q_{\text{local}}) = 0$ for $q \neq 0$, and the free residual perturbation Δ has bidegree $(1, 0)$. Suppose in addition that the chosen SDR is compatible with the local grading, so that s has bidegree $(0, -1)$ and j, p are supported in local degree zero. Then*

$$\boxed{Q_H = p\Delta j}, \quad (126)$$

all higher HPL terms vanish, and the local-to-residual spectral sequence collapses to

$$H^p(\mathfrak{g}; H^0(q_{\text{local}})) . \quad (127)$$

Proof. The homotopy s lowers local degree by one. Since both $j(H)$ and the image retained by p lie in local degree zero,

$$p\Delta(s\Delta)^r j = 0 \quad (r \geq 1). \quad (128)$$

The HPL series therefore reduces to (126). The E_1 page has only the row $q = 0$, so no higher differential can enter or leave it. $\square$

The proposition concerns the perturbation obtained from the *free metric BV bicomplex*. It must not be inferred by naively restricting a nonlinear minimal Q -manifold. In the Dneprov–Grigoriev variables, curvature-dependent terms schematically of the form $\xi\xi W$ and $\xi\xi C$ occur in ghost transformations and can have total compact degree zero [19]. They belong to the nonlinear comparison and must either be identified with the interacting deformation or removed by the equivalence back to the free metric presentation before Proposition 8.4 is applied.

Proposition 8.5 (Formal filtered-transfer criterion). *Suppose the zero-mode split produces (121); the filtration is convergent; $H^q(q_{\text{local}})$ vanishes in every row that can enter or leave $(4, 0)$; and the transferred differential on $H^0(q_{\text{local}})$ is the strict residual Chevalley–Eilenberg differential, with all relevant $Q_{[k]}$ absent. Then the full cohomology in the selected filtered degree is isomorphic, as a vector space, to the residual group*

$$H^4(\mathfrak{g}; H^0(q_{\text{local}})) . \quad (129)$$

If in addition the retract is compact-degree equivariant and cyclic, the Cartan homotopy and the cohomological Hermitian pairing transfer, and the dressed inclusion is the isometry (115).

Proof. Under the stated row and strictness assumptions no differential (125) enters or leaves $(4, 0)$, so the bounded spectral sequence collapses there to its E_2 value. The equivariant and cyclic statements are (112) and (115). The cyclic splitting fixes the extension of the pairing from the associated graded object. $\square$

8.6 End-to-end centered integration in the polynomial category

The transferred coefficient matrices of Proposition 8.3 can be coupled directly to the independently generated residual BFV/CE ghosts. This removes the hand-specified $E/A/L$ matrices from the final centered rank calculation.

Computational Proposition 8.6 (Metric-to-residual integration). *In the parity-complete algebraic polynomial complex, the pipeline*

$$\text{metric BV rows} \longrightarrow H(q) \longrightarrow C^\bullet(\mathfrak{so}(4, 2); H(q)) \quad (130)$$

has the exact centered dimensions and ranks

	$\dim C^3$	$\dim C^4$	$\dim C^5$	$\text{rank } d_3$	$\text{rank } d_4$	$\dim H^4$
<i>vacuum</i>	147	407	833	116	291	0
<i>one particle</i>	580	2622	7314	520	2102	0
<i>two particles</i>	0	55	385	0	53	2.

(131)

For the two-particle row, the next space has dimension 1155 and $d_5 d_4 = 0$ exactly. Orientation reversal acts on the two kernel vectors by

$$\text{diag}(-1, +1). \quad (132)$$

Pulling the positive energy-two electric/magnetic curvature form back to the metric representatives gives the raw kernel Gram matrix

$$\text{diag}\left(\frac{5}{64}, 5\right). \quad (133)$$

After exact normalization and multiplication by the intrinsic unit residual ghost norm, the representative Gram matrix is I_2 .

The odd and even eigenvectors are respectively the Pontryagin and Weyl-square directions. The calculation verifies the vector-space transfer and positive representative form in this algebraic model. It does not by itself identify the curvature-coordinate normalization with the complete cross-energy local BV/BFV symplectic pairing. That final cyclic normalization comparison remains visible below. The certificate and machine-readable ranks are produced by `symbolic/verify_conformal_metric_to_residual_integration.py`.

8.7 Two remaining field-theoretic hypotheses

The smooth BGG theorem and Theorem 2.5 remove the global PDE-surjectivity and finite-mode preimage gaps. Propositions 8.3 and 8.6 additionally close the kernel, image, noncompact-action, and centered-rank calculation in an explicit polynomial metric-BV model. Identifying that model and its representative form with the complete field-theoretic BV/BFV complex is concentrated in two statements.

Field-Theoretic Hypothesis 8.7 (Strict pure-Weyl transfer hypotheses). The gauge-fixed free strict-pure-Weyl BV complex has the following properties in the cylinder problem under consideration.

1. *Field-theoretic BV and cyclic identification.* The fifteen global reducibilities and their genuine dual constraint/antifield sectors can be split without double counting. After non-minimal doublets are contracted, the complete nonzero-mode free local BV complex is quasi-isomorphic, in the selected cylinder window, to the raw polynomial detour rows of (116), with no additional local or antifield row entering or leaving the centered degree. Its cross-energy BV/Krein pairing admits a cyclic transfer whose normalization agrees with the curvature form used in Proposition 8.6.
2. *Residual BFV choice.* The selected closed-universe boundary problem gauges all fifteen residual generators and its BFV pairing induces the compact-energy four-ghost polarization and complementary-degree determinant insertion of Lemma 6.1, including the common normalization (90) and the parity-compatible real structure.

These are not consequences of the residual rank matrices. The raw calculation already contains the minimal detour ghost and antifield rows and proves homotopy-equivariant noncompact transfer; the first item asks for the remaining identification with the complete field-theoretic BV domain and its cross-energy cyclic form. The second is stronger than verifying the natural Berezin saturation inside the already chosen residual complex.

Corollary 8.8 (Conditional algebraic strict-pure-Weyl result). *If Field-Theoretic Hypothesis 8.7 holds, the complete free classical local-plus-residual BV cohomology in the algebraic cylinder category and selected degree is (91). Its transferred cohomological Hermitian form is I_2 , and its local Euler–Lagrange quotient is (105) with form I_1 .*

Proof. Theorems 2.4 and 2.5 identify the one-particle coefficient module. Proposition 8.1 gives its free Fock lift. Propositions 8.3 and 8.6 give the strict induced residual action and centered cohomology in the algebraic polynomial model. Item 1 identifies this model and its cyclic form with the complete local BV complex. Item 2 fixes the selected ghost pairing and normalization. Equation (115) then identifies the residual representative Gram matrix with the transferred pure-Weyl BV cohomological form. $\square$

The global smooth BGG theorem, curvature isomorphism (23), action factorization (9), raw metric-BV transfer, and integrated centered ranks are exact inputs. The two hypotheses above state the remaining obligations without folding the cross-energy cyclic pairing or BFV boundary choice into the metric-to-curvature theorem. A Hilbert, Krein, distributional, or completed Fock statement would require separate continuity, closed-range, and convergence results.

9 Scope, interpretation, and open questions

Residual gauge choice. The result is specific to quotienting the full residual $SO(4, 2)$ action on the compact cylinder. If some generators are retained as global or asymptotic charges, the physical complex is different. In particular, Cartan contraction cannot be used to erase nonzero D energy when D is the physical Hamiltonian.

No positive graviton Fock claim. The unreduced module remains indefinite and the one-particle residual cohomology is zero. The positive classes are weight-four scalar vertices. Calling I_2 a positive graviton Hilbert space would therefore misidentify both the cohomological degree and the state–operator map.

Classical and free. The result concerns the free classical complex and its finite residual reduction. It does not show that the interaction preserves the reduced space or pairing. Nor does it address the quantum master equation. A local Diff $\times$ Weyl anomaly could obstruct quantum nilpotency even if Corollary 8.8 is fully established.

Vertex versus dynamics. The two residual classes do not define two independent local gravitational dynamics. The parity-even class is the Weyl action direction; the parity-odd class is a theta direction, locally related by a canonical boundary transformation. The positive dynamical quotient is therefore one-dimensional.

The remaining classical bridge. The all-level smooth metric-to- $\mathcal{W}_{\text{geom}}$ identification and the algebraic $E/A/L$ curvature intertwiner are exact. The raw polynomial metric-BV complex now supplies the antifield-row inventory, explicit noncompact homotopies, strict induced conformal action, and end-to-end centered ranks. Two sharper obligations remain: identify its curvature-coordinate form with the complete cross-energy cyclic local-BV pairing, and derive the chosen four-ghost polarization and normalized pairing from the residual BFV boundary problem. The

direct Taub argument does not require first identifying a degree-three BGG representative, although that geometric comparison remains valuable. Extension to a chosen Hilbert or Krein completion is a separate analytic question. The exact ranks in Table 1 remain independent convention and regression checks.

10 Conclusion

The conventional free oscillator description of pure conformal gravity on $\mathbb{R} \times S^3$ is indefinite. Residual conformal reduction changes the question rather than re-signing those oscillators. The absolute residual complex carries a Cartan contraction which eliminates every noncentered compact degree. At the ghost number four selected by the centered polarization this turns the apparent infinite problem into the three sectors $N = 0, 1, 2$. The first two are acyclic; the last has exactly two classes and positive Gram matrix:

$$H_{\text{res,min}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.$$

Descent identifies them as the Weyl-square and Pontryagin vertex directions, and the quotient by variationally trivial bulk functionals leaves one positive dynamical direction.

The result is both stronger and narrower than a sampled positivity calculation. It is exhaustive for the stated minimal residual complex, and the same centered ranks now arise from an independent raw polynomial metric-BV transfer, but it concerns vertices rather than propagating particles. Its promotion to the complete field-theoretic pure-Weyl BV/BFV theory, including the cyclic normalization and boundary choice, is controlled by the explicit hypotheses of Corollary 8.8. Quantum consistency and interaction descent are separate questions.

A Conventions

The cylinder radius is one and the Lorentzian signature is $(-, +, +, +)$. Compact energy is the eigenvalue of $D = i\partial_t$ in the oscillator convention. The compact algebra is $SO(4) \cong SU(2)_L \times SU(2)_R$. A label (j_L, j_R) has dimension $(2j_L + 1)(2j_R + 1)$. Positive chirality means the lowest Weyl primary $(2; 2, 0)$; parity exchanges j_L and j_R .

The normalized module convention is (44). The reduced curvature action (8) differs by an overall factor and sign from the common $-\int C^2$ oscillator convention. All moment-map comparisons derive the symplectic form and Taub kernel from the same action, so the overall action factor cancels. Lorentzian self-duality may insert a factor of i in the odd parity combination; no conclusion depends on that convention.

The residual generator degrees are $(0^7, -1^4, +1^4)$ and dual ghost degrees are their negatives. Ghost number and compact degree are independent. The selected absolute residual degree in this paper is ghost number four, fixed by (63); it is not ordinary ghost number zero. We call it a candidate physical degree only under the full residual gauging specified in Section 5.2.

B Exact rank certification

The one-particle matrices contain radicals. They are represented over

$$K = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}). \tag{134}$$

For a lower bound on rank, the calculation maps this field to $\mathbb{F}_{241}$ by

$$i \mapsto 64, \quad \sqrt{2} \mapsto 22, \quad \sqrt{3} \mapsto 56, \quad \sqrt{5} \mapsto 103. \quad (135)$$

Each image obeys its defining square relation. A nonzero minor after this good-prime reduction is a nonzero characteristic-zero minor. Exact nilpotency provides the complementary upper bound

$$\text{rank } d_{q-1} + \text{rank } d_q \leq \dim C^q. \quad (136)$$

For the one-particle complex the modular lower bounds saturate this inequality. For the two-particle complex the two explicit independent kernel vectors provide the upper bound which the modular rank saturates. Thus the ranks quoted in Section 6 are exact characteristic-zero ranks, not probabilistic modular estimates.

C Claim-status and verification index

The residual computation used in the first draft was frozen at repository commit

`e928f257c25099eb534eb34109dfc1dc6a3127a1`.

The global BGG bridge was added in the subsequent theorem revision based at

`c471b99f5e3708e692b1c25238f6272c9e29b48f`.

The companion snapshot records both layers, the environment, hashes, expected-failure guards, and command lines. The complete paper verification is run with

`python3 symbolic/verify_conformal_paper_free.py`

Claim	Status	Evidence / boundary
Formal pure-Weyl detour identities	Exact	Branson–Gover theorem plus action-normalized factorization and finite scalar-block audits.
Global smooth deformation cohomology and curvature isomorphism	Exact smooth theorem	The flat BGG fine resolution and $H^\bullet(\mathbb{R} \times S^3)$ give $H_{\text{def}}^1 = H_{\text{def}}^2 = 0$ and hence (23). This does not establish a completed Hilbert/Krein isomorphism or the BV/BFV zero-mode split.
Euclidean polynomial detour ranks, degrees 2–6	Exact finite certificate	Rational matrices verify kernels, ranks, and the 15 residual zero modes; they are an independent convention audit rather than the source of the global theorem.
Algebraic cylinder realization and stable generator coefficients	Exact symbolic theorem	Full coordinate Weyl/Bach calculation at symbolic n , nonzero $E/A/L$ pivots, explicit same-block metric preimages, character exhaustion, invariant form, and cutoff-stable conformal brackets.
Residual Cartan homotopy reduction	Theorem	Explicit homotopy ι_D/δ on every nonzero total degree; this is a quotient only in the full residual-gauging problem.
Centered vacuum and one-particle sectors	Exact	Semisimple Lie-algebra cohomology for the vacuum; cutoff-complete absolute rank certificate for one particle.
Centered two-particle sector and $G_{\text{res}} = I_2$	Exact residual theorem	Independent relative and absolute calculations; empty incoming absolute space and exact rank 53.
Vertex descent and dynamical/topological split	Exact locally	Weight-four descent, Chern–Weil transgression, and Euler–Lagrange quotient. Global theta effects are retained.
Full local-plus-residual pure-Weyl BV result	Conditional	Requires the two parts of Field-Theoretic Hypothesis 8.7: full equivariant cyclic BV transfer with row control, and the selected residual BFV polarization and normalization.
Interpretation as a physical state space	Choice-dependent	Requires a boundary and gauge principle that treats all fifteen generators, including D , as gauge rather than global charges.
Quantum anomaly freedom and interacting probability	Not claimed	Outside the scope of this paper.

References

- [1] T. Branson and A. R. Gover, “The conformal deformation detour complex for the obstruction tensor,” [arXiv:math/0605192](#).
- [2] A. R. Gover and L. J. Peterson, “The ambient obstruction tensor and the conformal deformation complex,” *Pacific J. Math.* **226** (2006) 309–351, [arXiv:math/0408229](#).
- [3] A. R. Gover, P. Somberg, and V. Souček, “Yang–Mills detour complexes and conformal geometry,” *Commun. Math. Phys.* **278** (2008) 307–327, [arXiv:math/0606401](#).
- [4] M. Benini, G. Musante, and A. Schenkel, “Green hyperbolic complexes on Lorentzian manifolds,” *Commun. Math. Phys.* (2023), [doi:10.1007/s00220-023-04807-5](#), [arXiv:2207.04069](#).
- [5] A. Čap, “Overdetermined systems, conformal geometry, and the BGG complex,” in *Symmetries and Overdetermined Systems of Partial Differential Equations*, IMA Vol. Math. Appl. **144** (Springer, 2008) 1–24, [arXiv:math/0610225](#).

- [6] A. Čap, “Infinitesimal automorphisms and deformations of parabolic geometries,” *J. Eur. Math. Soc.* **10** (2008) 415–437, [arXiv:math/0508535](#).
- [7] A. Čap and K. Hu, “Bounded Poincaré operators for twisted and BGG complexes,” *J. Math. Pures Appl.* **179** (2023) 253–276, [arXiv:2304.07185](#).
- [8] K.-j. Hamada and S. Horata, “Conformal algebra and physical states in non-critical 3-brane on $\mathbb{R} \times S^3$,” *Prog. Theor. Phys.* **110** (2003) 1169–1210, [arXiv:hep-th/0307008](#).
- [9] K.-j. Hamada, “Building blocks of physical states in a non-critical 3-brane on $\mathbb{R} \times S^3$,” *Int. J. Mod. Phys. A* **20** (2005) 5353–5398, [arXiv:hep-th/0402136](#).
- [10] K.-j. Hamada, “Conformal field theory on $\mathbb{R} \times S^3$ from quantized gravity,” *Int. J. Mod. Phys. A* **24** (2009) 3073–3110, [arXiv:0811.1647](#).
- [11] K.-j. Hamada, “BRST analysis of physical fields and states for 4D quantum gravity on $\mathbb{R} \times S^3$,” *Phys. Rev. D* **86** (2012) 124006, [arXiv:1202.4538](#).
- [12] S. M. Kuzenko and M. Ponds, “Conformal geometry and (super)conformal higher-spin gauge theories,” *JHEP* **05** (2019) 113, [arXiv:1902.08010](#).
- [13] M. Beccaria, X. Bekaert, and A. A. Tseytlin, “Partition function of free conformal higher spin theory,” *JHEP* **08** (2014) 113, [arXiv:1406.3542](#).
- [14] N. Boulanger and M. Henneaux, “A derivation of Weyl gravity,” *Annalen Phys.* **10** (2001) 935–964, [arXiv:hep-th/0106065](#).
- [15] G. Barnich, F. Brandt, and M. Henneaux, “Local BRST cohomology in gauge theories,” *Phys. Rept.* **338** (2000) 439–569, [arXiv:hep-th/0002245](#).
- [16] G. Barnich, F. Brandt, and M. Henneaux, “Local BRST cohomology in the antifield formalism: I. General theorems,” *Commun. Math. Phys.* **174** (1995) 57–92, [arXiv:hep-th/9405109](#).
- [17] G. Barnich, M. Henneaux, T. Hurth, and K. Skenderis, “Cohomological analysis of gauged-fixed gauge theories,” *Phys. Lett. B* **492** (2000) 376–384, [arXiv:hep-th/9910201](#).
- [18] R. R. Metsaev, “BRST–BV approach to conformal fields,” *J. Phys. A* **49** (2016) 175401, [arXiv:1511.01836](#).
- [19] I. Dneprov and M. Grigoriev, “Presymplectic BV–AKSZ formulation of conformal gravity,” [doi:10.1140/epjc/s10052-022-11082-6](#), [arXiv:2208.02933](#).
- [20] G. Barnich, X. Bekaert, and M. Grigoriev, “Notes on conformal invariance of gauge fields,” *J. Phys. A* **48** (2015) 505402, [arXiv:1506.00595](#).
- [21] E. Altas and B. Tekin, “Linearization instability for generic gravity in AdS,” *Phys. Rev. D* **97** (2018) 024028, [arXiv:1705.10234](#).
- [22] E. Altas and B. Tekin, “Second order perturbation theory in general relativity: Taub charges as integral constraints,” *Phys. Rev. D* **99** (2019) 104078, [arXiv:1903.11982](#).
- [23] C. Chevalley and S. Eilenberg, “Cohomology theory of Lie groups and Lie algebras,” *Trans. Amer. Math. Soc.* **63** (1948) 85–124.

- [24] W. Fischler and S. Kundu, “Physical effects of the gravitational Θ -parameter,” JHEP **05** (2017) 098, [arXiv:1612.06010](#).